

Johnny Adrián García Betancourt

M.Sc. Student in Electrical Engineering • Control & Robotics, KFUPM

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Research Profile

Mechatronics engineer and M.Sc. researcher studying the geometric, dynamical, and energetic structure of nonlinear physical systems and how that structure can guide machine-learning models. My work combines differential geometry, port-Hamiltonian modelling, passivity theory, differential flatness, and deep learning for physics-guided representation learning and fault diagnosis. I aim to extend this structural viewpoint toward Scientific Machine Learning and Scientific AI, developing learning methods that incorporate physical laws, geometry, symmetries, and energy structure into their representations and models.

Education

M.Sc. Student in Electrical Engineering – Control & Robotics Track 2025–Present

King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia

GPA: 3.938/4.00

Thesis: *Bridging Physical Safety and Active Tolerance in Robotic Manipulators: A Hierarchical Energy-Aware Architecture.*
Expected defense: December 2026–January 2027.

Selected Graduate Coursework: Linear Control Systems (A+), Optimal Control Theory & Applications (A+), Classical Mechanics (PHYS 507, A+), Advanced Methods for Theoretical Physics (PHYS 571, A+), Image Processing (A+), Special Topics in Intelligent Automation & Robotics (A+), Robotics & Control (A).

Graduate Coursework in Automation and Control (Master's Program)

2022–2024

Escuela Superior Politécnica del Litoral (ESPOL), Ecuador

Certified Coursework Avg: 9.97/10

Completed advanced graduate coursework with near-perfect marks; transferred to KFUPM prior to thesis completion.

Selected Coursework: Machine Learning-Based Control (100/100), Identification & Control of Dynamic Systems (100/100), Embedded Control Systems (100/100), Industrial Robotics & Computer Vision (97/100), Industrial Automation Systems (100/100).

B.Sc. in Mechatronics Engineering (Engineer in Mechatronics)

2017–2022

Escuela Superior Politécnica del Litoral (ESPOL), Ecuador

Graduation Date: Dec 7, 2022

Thesis: *Mechanical and Electrical Design, and Automation of an Industrial Liquid Filling Machine* (Thesis Grade: 9.20/10).
Awarded Best Mechatronics Thesis.

Research Experience

Spectral-Hamiltonian Algebraic Power Evaluation (SHAPE Framework)

Research Project | KFUPM

Conceived and formulated an observer-free, physics-guided fault detection and isolation methodology interfacing port-Hamiltonian energy balance ($\dot{H} \leq y^T u$) with differential flatness. Formulated the algebraic port-power residual that computes nominal power trajectories directly from reference profiles, eliminating online observer synthesis and numerical differentiation. Led the multi-body physical simulation and trajectory data-generation pipeline on a 6-DOF Kinova Gen3 manipulator. Transitioned the methodology to native numerical tensors (multichannel temporal, CWT time-frequency scalograms, and FFT spectral tensors), co-designed a shared-encoder multi-joint CNN with late fusion, and personally implemented the training, evaluation, and benchmarking pipeline in Python/PyTorch, achieving a 91.7% macro F1-score across nominal and degraded operating regimes.

Hierarchical Energy-Aware Architecture for Safe Robotic Interaction

M.Sc. Thesis | KFUPM

Developing a hierarchical control framework bridging deterministic physical passivity with active fault tolerance in robotic manipulators. Formulated a low-level Port-Hamiltonian Energy Tank supervisor to enforce strict passivity ($\dot{H} \leq 0$) and bound energy injection under actuator Loss of Effectiveness (LoE). Developed a coordinate-free high-level trajectory planner parametrizing end-effector pose directly on the Lie group $SE(3)$ along with kinematic self-motion for 7-DOF redundant manipulators, ensuring dynamic feasibility without coordinate singularities.

Active Fault-Tolerant Flatness-Based Model Predictive Control

Research Manuscript | Under Review

Developed an active fault-tolerant model predictive control architecture for nonlinear robotic manipulators subject to progressive actuator degradation. The formulation algebraically maps shrinking physical actuator torque limits into state-dependent acceleration constraints in flat output space, eliminating numerical chattering, reducing energy consumption by 66% (from 17,064 to 5,668 N²m²s), and accelerating trajectory optimization to real-time execution speeds (0.01 s).

Geometric Modelling and Invariant State Estimation

Coursework & Research Projects | KFUPM

Investigated geometric mechanics and symmetry-preserving estimation for underactuated nonlinear dynamical systems. Developed multi-body dynamic models for an underactuated underwater vehicle actuated by internal shifting mass and variable-buoyancy piston systems. Implemented closed-loop simulation and Kalman filtering state estimation architectures in MATLAB, Simulink, and Simscape Multibody, analyzing state reconstruction accuracy under measurement noise and unmodeled hydrodynamics as a foundation for invariant and structure-preserving learning.

Research Manuscripts & Working Papers

[1] **J. Garcia Betancourt** and N. Salguero España, “Active Fault-Tolerant Flatness-Based MPC for Robotic Manipulators under Parametric Mismatch and Actuator Degradation,” *Research Manuscript*, revised for resubmission, 2026.

[2] **J. Garcia Betancourt**, N. Salguero España, N. Iqbal, B. Brahmi, and R. Rashad, “Energy-Based Fault Detection and Isolation Framework for Flat Port-Hamiltonian Mechanical Systems,” *Research Manuscript*, in final preparation, 2026.

[3] U. I. Castilla Batun and **J. A. Garcia Betancourt**, “Techno-Economic Feasibility of Wind-Powered Hydrogen Production in Saudi Arabia: A Multi-Regional Assessment,” *Conference Manuscript*, revised for resubmission, 2026.

Technical Expertise

Scientific ML & Deep Learning	Physics-guided representation learning; custom 1D/2D CNNs, shared-weight multi-joint encoders and late feature fusion; temporal, CWT time-frequency and FFT/spectral representations; comparative representation benchmarking; multi-seed evaluation, clustered bootstrap uncertainty quantification and robustness analysis; SVMs, PCA and t-SNE latent-space analysis.
Mathematical Foundations & Differential Geometry	Linear and tensor algebra, multivariable calculus, ordinary and partial differential equations, Fourier and Laplace methods, numerical methods, probability and statistics, calculus of variations, constrained and convex optimization; smooth manifolds, tangent and cotangent bundles, differential forms, Lie brackets, Riemannian metrics, Lie groups and Lie algebras, exponential maps, and differential-geometric modelling.
Analytical Mechanics & Energy-Based Physical Modelling	Variational, Lagrangian and Hamiltonian mechanics; symplectic and Poisson formulations, conservation laws and symmetries, rigid-body and multibody mechanics; port-Hamiltonian systems, Dirac structures, passivity, energy shaping, energy tanks, and power/energy-based modelling.
Nonlinear Systems, Control & Optimization	Linear and nonlinear dynamical systems, state-space modelling, equilibrium and stability analysis, Lyapunov stability, LaSalle invariance, controllability and accessibility, geometric control, differential flatness, optimal control, MPC, Quadratic Programming (QP), trajectory optimization, inverse dynamics, and active fault-tolerant control.
State Estimation, Signal & Image Processing	Bayesian estimation, Lie-group and invariant state estimation, SLAM foundations, Fourier and spectral analysis, digital filtering, Continuous Wavelet Transform (CWT), spatial and frequency-domain image processing, feature extraction, pattern classification, and computer-vision foundations.
Scientific Computing & Engineering Tools	Python, PyTorch, MATLAB, Simulink, Simscape Multibody, Git, $\LaTeX$, CAD/CAE (SolidWorks, Inventor, ANSYS), instrumentation, PLC/SCADA systems, and industrial communication protocols.
Research interests: <i>scientific foundation models, structure-preserving learning, geometric deep learning, and inverse problems.</i>	

Academic and Professional Experience

Graduate Teaching & Laboratory Assistant <i>King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia</i> Instructing undergraduate laboratory sessions in electronics and instrumentation, delivering hands-on guidance on circuit analysis, operational amplifiers, signal conditioning, and electronic measurement systems.	2025–Present
Teaching Assistant – Statistics and Probability <i>Escuela Superior Politécnica del Litoral (ESPOL), Ecuador</i> Assisted undergraduate engineering courses; conducted problem-solving recitations on probability distributions, parameter estimation, and statistical data analysis; evaluated student assignments.	2022
Undergraduate Research & Prototyping Assistant <i>Escuela Superior Politécnica del Litoral (ESPOL), Ecuador</i> Contributed to design and hardware prototyping for mechatronic machinery projects, assisting in CAD modeling, sensor interfacing, and PLC-based automated control routines.	2021–2022
Maintenance and Automation Coordinator <i>Detecuator S.A., Ecuador</i> Supervised automated operations, industrial instrumentation, and predictive maintenance protocols for manufacturing facilities; engineered 2D/3D CAD solutions for electro-mechanical system modernization.	2021–2024
Assistant Electromechanical Designer <i>SEOR S.A., Ecuador</i> Developed engineering blueprints, electrical control cabinets, and automation layouts for industrial machinery and packaging lines.	2023–2024

Honors and Awards

Graduate Full Scholarship , King Fahd University of Petroleum & Minerals (KFUPM)	2025–Present
Best Mechatronics Thesis Award (1st Place) , Escuela Superior Politécnica del Litoral	2022
Best University Thesis Award (2nd Place Institution-Wide) , ESPOL	2022
Academic Excellence Recognition (Top Score) , National University Admission Exam (SENESCYT)	2017

Languages

Spanish – Native	English – Advanced / Academic working proficiency
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